Vanessa Guerra Canedo

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Relevant work experience

Museum Technician - Project Lead, USFWS-NMNH Biorepository | Smithsonian Institution, National Museum of Natural History (NMNH) | Washington, DC **08/2025 - Present**
Hours per week: 40

Coordinate development of a centralized freshwater mussel biorepository across Smithsonian teams, USFWS National Fish Hatcheries, and conservation partners.

Manage specimen curation, accessioning into EMu and FreezerPro, DNA isolation, and metadata; perform QA/QC and contribute to optimization of DNA and tissue storage for future genomic research.

Provide bioinformatics and computational genomics support to a museum researcher, including phylogenomic analyses, 3RAD-seq workflow development, and development of a long-read sequencing pilot.

Postdoctoral Fellow | National Institute of Allergy and Infectious Disease (NIAID), Vaccine Research Center | Bethesda, MD **03/2022 - 06/2025**
Hours per week: 40

Adapted high-throughput sequencing (HTS) for broad application across influenza A viruses, developing wet-lab and computational workflows for single- and whole-genome sequencing of low-viral-load samples. Improved variant detection to 0.5% using Unique Molecular Identifiers (UMIs) and integrated deep-learning tools into PacBio analysis, using synthetic DNA controls to quantify error rates and validate improvements in sequence accuracy. Used Python, R, Perl, Snakemake, Unix/Linux/Bash, and High-Performance Computing (HPC).

Provided sequencing, bioinformatics, and computational genomics support to Vaccine Research Center (VRC) influenza teams for highly pathogenic avian influenza (H5Nx) response and studies of vaccines, antibodies, and viral evolution; contributed to publications in Science, Nature Microbiology, and Nature Communications.

Trained VRC and international researchers in NGS and bioinformatics, including a scientist from South Africa's National Institute for Communicable Diseases in applying HTS to influenza; delivered the VRC NGS lecture in 2024 and 2025.

Research Associate | Department of Invertebrate Zoology, National Museum of Natural History, Smithsonian Institution | Washington, DC **09/2018 - 10/2024**
Hours per week: 10

Conducted comparative and evolutionary genomics analyses supporting doctoral research using Smithsonian bioinformatics and HPC resources.

Scientist | MilliporeSigma | Rockville, MD **08/2021 - 03/2022**
Hours per week: 40

Provided scientific oversight for high-throughput, agnostic NGS workflows for adventitious agent detection in biomanufacturing within a GMP environment. Performed sequencing data analysis and prepared final technical reports; reviewed batch records and results for assay performance, QA/QC, RFT, and regulatory compliance. Managed specimen accessioning, chain of custody, and LIMS records for data integrity and traceability, collaborating across teams to optimize standardized sequencing workflows.

Postdoctoral Fellow | University of California, Merced | Merced, CA **01/2021 - 06/2021**
Hours per week: 40

Coordinated startup and implementation of a 12-species population and community genomics project for the California Conservation Genomics Consortium, including field collections and biological sample acquisition. Co-instructed a graduate genomics data analysis course at UC Davis using Git/GitHub for computational instruction and workflow sharing; planned molecular and NGS work was limited by COVID-19-related supply shortages.

Research Assistant (Doctoral Researcher) | Simon Fraser University | Burnaby, BC Canada **09/2015 - 12/2020**
Hours per week: 40

Investigated the evolution of reproductive genes and strategies in sea stars, with related work on human reproductive genes. Developed and used bioinformatic/genomic data pipelines to process, assemble, visualize, and analyze genomic and transcriptomic data using R, Perl, Bash, Unix/Linux, and High-Performance Computing (HPC), including genome assembly, genome annotation, phylogenetic analysis, and functional analysis. Integrated genomic and transcriptomic data to resolve highly similar reproductive genes and established quantitative expression thresholds to identify potential misidentifications; pipeline development included technical review from bioinformaticians and evolutionary biologists.

Intern | Smithsonian Conservation Biology Institute (SCBI), National Zoological Park, Smithsonian Institution | Washington, DC **05/2019 - 06/2020**
Hours per week: 40

Performed computational genomic analyses supporting vertebrate conservation genomics research.

Laboratory Manager | Wake Forest University | Winston-Salem, NC **09/2014 - 07/2015**
Hours per week: 40

Managed a molecular biology laboratory supporting fish and oyster genomics research and adapted 2b-RAD-seq workflows from DNA extraction and PCR through NGS library preparation and sequencing data analysis.

Standardized molecular protocols and trained laboratory personnel in molecular biology and NGS methods.

Developed and delivered an NGS workshop integrating wet-lab and bioinformatics methods.

Intern | Smithsonian Environmental Research Center | Tiburon, CA — **12/2009 - 11/2010**
Hours per week: 40

Supported biodiversity monitoring, specimen identification, and ecological data management.

Marine Laboratory Technician | Telonicher Marine Laboratory, Humboldt State University | Trinidad, CA — **09/2008 - 12/2009**
Hours per week: 12

Maintained collections and supported specimen identification and laboratory operations.

Education, certification or licenses

Simon Fraser University | Burnaby, BC Canada — **Completion date:** 12/2020
Doctorate degree
Major: Biology

San Francisco State University | San Francisco, CA — **Completion date:** 12/2014
Master's degree
Major: Biology (Ecology, Evolution, and Conservation Biology)

Humboldt State University | Arcata, CA — **Completion date:** 12/2009
Bachelor's degree
Major: Bachelor of Science | **Minor:** Scientific, Recreation or Leadership Diving

Job-related training

Data Carpentry Instructor, The Carpentries (2018); Best Practices for Using NGS-Based Datasets to Detect Positive Selection and Convergent Evolution, Society for Integrative & Comparative Biology (2018–2019); Comparative Genomics, Simon Fraser University (2015); NGS Techniques, Duke University (2014); Genome Evolution and Annotation, San Francisco State University (2013); Genetics for Conservation, Red de Genética de la Conservación (2012).

Language skills

English — **Spoken:** Advanced | **Written:** Advanced | **Read:** Advanced

Spanish — **Spoken:** Advanced | **Written:** Advanced | **Read:** Advanced

Professional publications

Abu-Shmais AA, Freeman G, Creanga A, Vukovich MJ, Malla T, et al., Guerra Canedo V, et al. Cross-neutralizing and potent human monoclonal antibodies against historical and emerging H5Nx influenza viruses. Nature Microbiology. 2025;10(11):2903–2918.

Kanekiyo M, Gillespie RA, Cooper K, Guerra Canedo V, et al. Pre-exposure antibody prophylaxis protects macaques from severe influenza. Science. 2025;387(6733):534–541.

Ko SH, Radecki P, Belinky F, et al., Guerra Canedo V, et al. Rapid intra-host diversification and evolution of SARS-CoV-2 in advanced HIV infection. Nature Communications. 2024;15:7240.

Houck ML, Koepfli KP, et al., Guerra V, et al. Chromosome-length genome assemblies and cytogenomic analyses of pangolins reveal remarkable chromosome counts and plasticity. Chromosome Research. 2023;31(2):13.

Hart MW, Guerra VI, Allen JD, Byrne M. Cloning and selfing affect population genetic variation in simulations of outcrossing sexual sea stars. The Biological Bulletin. 2021;241(3):286–302.

Guerra VI, Haynes G, Byrne M, Hart MW. Selection on genes associated with the evolution of divergent life histories: Gamete recognition or something else? Evolution & Development. 2021;23(5):423–438.

Guerra VI, Haynes G, Byrne M, et al. Nonspecific expression of fertilization genes in the crown-of-thorns Acanthaster cf. solaris: Unexpected evidence of hermaphroditism in a coral reef predator. Molecular Ecology. 2020;29(2):363–379.

Hart MW, Stover DA, Guerra V, Mozaffari SV, Ober C, Mugal CF, Kaj I. Positive selection on human gamete-recognition genes. PeerJ. 2018;6:e4259.

Hart MW, Guerra VI. Finding genes and lineages under selection in speciation. Molecular Ecology. 2017;26(14):3587–3590.

Sangesland M, Gillespie RA, Cooper K, Ellis D, Boyoglu-Barnum S, Guerra Canedo V, et al. A mosaic nanoparticle displaying influenza hemagglutinin stem of multiple subtypes elicits broadly protective immunity. Science Translational Medicine. Submitted, 2026.

Pfeiffer J, Hampson E, Guerra Canedo V, Goetz F, Keogh S, et al. The evolution of cementation in freshwater mussels (Bivalvia: Unionoida). Zoological Journal of the Linnean Society. Submitted, 2026.